CEAS Co-op Reflection System: D2 Detailed Design

Header, conventions, and D2 overview

Project title: CEAS Co-op Reflection System

Goal statement: Let CEAS co-op students reflect on each completed rotation using rotation-specific questions, give assigned advisors a clear summary with workplace supervisor observations, and keep each reflection private to the student and their assigned advisor.

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Scope: D2 finishes every component that D1 deferred (C1, C3, C4, C5, C6) and the glue components (sign-in, logging, configuration, notifications). It adds the edge-case table, traceability table, design rationale, and peer review notes. C2 and C7 were detailed in D1, so D2 only refers back to them.

Conventions

Every figure repeats the project title and goal statement at its top. Each figure's own key sits below it as well.

- **Figure 1 (state machine):** A rounded box is a state of the student's reflection page. A solid arrow is a normal transition, labeled "event [condition]". A dashed arrow is an error path, labeled the same way. A box with a dashed outline is an error state. A filled dot is the start, and a box with a thick outline is where the flow ends.
- **Figure 2 (block diagram):** A solid box is a component we build. A dashed box is an external system (E1 sign-in, E2 University Records). A cylinder is a persistent store. An arrow points from the caller to the one being called, and its label is an interface ID. Glue components carry the word "glue" in their box.
- **IDs:** Interface IDs (I1 to I13) keep their D0 meaning. EC-01 and up are edge cases, OQ-1 and up are open questions, and R1 and up are rationale entries. Requirement IDs (US, UC, AC) are from Assignment 4. DR-1 to DR-6 and NFR-1 to NFR-5 are the draft rules and non-functional requirements, numbered in the traceability table.
- **Errors:** REST errors use an HTTP status plus a lowercase code, for example 403 access_denied. Calls that are not HTTP (database and in-process calls) use the code alone.

Overview in two minutes

The CEAS Co-op Reflection System lets a CEAS student write a reflection after each co-op rotation. It shows an assigned advisor a short summary with the workplace supervisor's observations, and it keeps each reflection private to the student and that advisor. Students, advisors, and supervisors all use one web front end (C1), which calls three services: the Reflection Service (C3), the Access Control Service (C2), and the Supervisor Evaluation Service (C6). Prompts live in C4, records live in C7, sign-in comes from the university (E1), and enrollment, rotation, advisor, and supervisor facts come from University Records (E2).

D1 detailed the two components the product depends on most. C7 got a relational data model with eight tables, and C2 got the owner-or-assigned-advisor access check, plus the save-once algorithm, the API contract for C2 and C7, and the technology choices.

D2 adds the other five components and the glue, each with a data model (or the words "no persistent data"), behavior, and an interface with errors. It also adds an error-path diagram, 40 edge cases, a traceability table for every requirement, nine rationale entries, and the peer review action items. It revises three D0 interfaces (I3, I4, I5) because they leaked internals or grew too many parameters.

Known gaps and open questions

ID	Open question or gap	Assumption this design makes until it is answered
OQ-1	How do workplace supervisors sign in? E1 is a university sign-in, and supervisors are outside the university.	CEAS issues supervisors a login that E1 accepts, or a one-time link. C6 only needs a verified supervisor_ref, so the design does not change.
OQ-2	Can a student have two current advisors?	Yes. The D1 model allows it, and both advisors can read.
OQ-3	How is a mistake in a submitted reflection or evaluation corrected?	Students are expected to resubmit the evaluation to include the correct responses.The design allows resubmission until the deadline, and the most recent submission replaces the earlier one.
OQ-4	Which rotation numbers are supported, and what is the exact prompt wording? The draft rules say CEAS must confirm both.	Rotations 1 to 3 have placeholder prompts. C4 holds them as data, so the real wording needs no code change.
OQ-5	What are the retention and deletion rules? The draft rules say they need separate requirements after university guidance.	The system never deletes data, and no deletion feature is designed. This is a stated gap (see DR-6).
OQ-6	May students see the supervisor's evaluation?	Yes, both students and the assigned advisor receive it (D1 API).
OQ-7	Does E2 offer a JSON API with the fields we need, and how fast is it?	Yes, with a 2-second timeout. If E2 only offers a nightly file, the local copy in C7 becomes the main source.

ID	Open question or gap	Assumption this design makes until it is answered
OQ-8	What is the wording and version of the data-use notice?	The notice is versioned text. C7 stores one acknowledgment per student per version.
OQ-9	Will UC IT approve the hosting platform for student data?	A managed platform is used for the demo only. Real student data waits for approval.
OQ-10	When can a supervisor submit an evaluation? D0 says "end of rotation."	End of rotation.
OQ-11	Should anyone receive email when something is submitted? No requirement asks for it.	No notifications are built (see the glue components).

Corrections to D0 and D1

- **D0, interface labels:** The block diagram has I8 and I9 swapped. The interface table and the component table agree that I8 is C2 to E2 and I9 is C3 to C7, so the diagram labels are the error. Fix: swap the two labels on the diagram.
- **D1, text fixes in the team's copy:** The data-size line must read 1,500 reflections and 6,000 answers now, and 150,000 and 600,000 at 100 times. The advisor-assignment window is 15 minutes everywhere, not 10 in one place. "Let D2 change ships" in the Alembic row should read "Lets D2 changes ship". The draft note and the license-check sentence come out before submission.
- **D0 interfaces I3, I4, I5:** The D0 text is unchanged. Section 2 gives the revised versions, and rationale entry R6 explains why.
- **D1, gap found while writing D2:** D1 gave an advisor no way to find a student. D2 adds GET /v1/advisors/me/students to C2.
- **D1, additive API changes:** the ?include=summary option on GET /v1/reflections/{reflection_id} and the roster endpoint add to the D1 API without changing it, which the D1 versioning rule says is not a breaking change.

Finished components, the D2 diagrams

Every component now has a data model, behavior, and an interface. Only C7 holds the system's records, and C4 holds the prompt wording in its own small store. Everything else keeps no persistent data. C2 and C7 are in D1, so this section covers the rest in order: C4, C5, C3, C1, C6, then the glue.

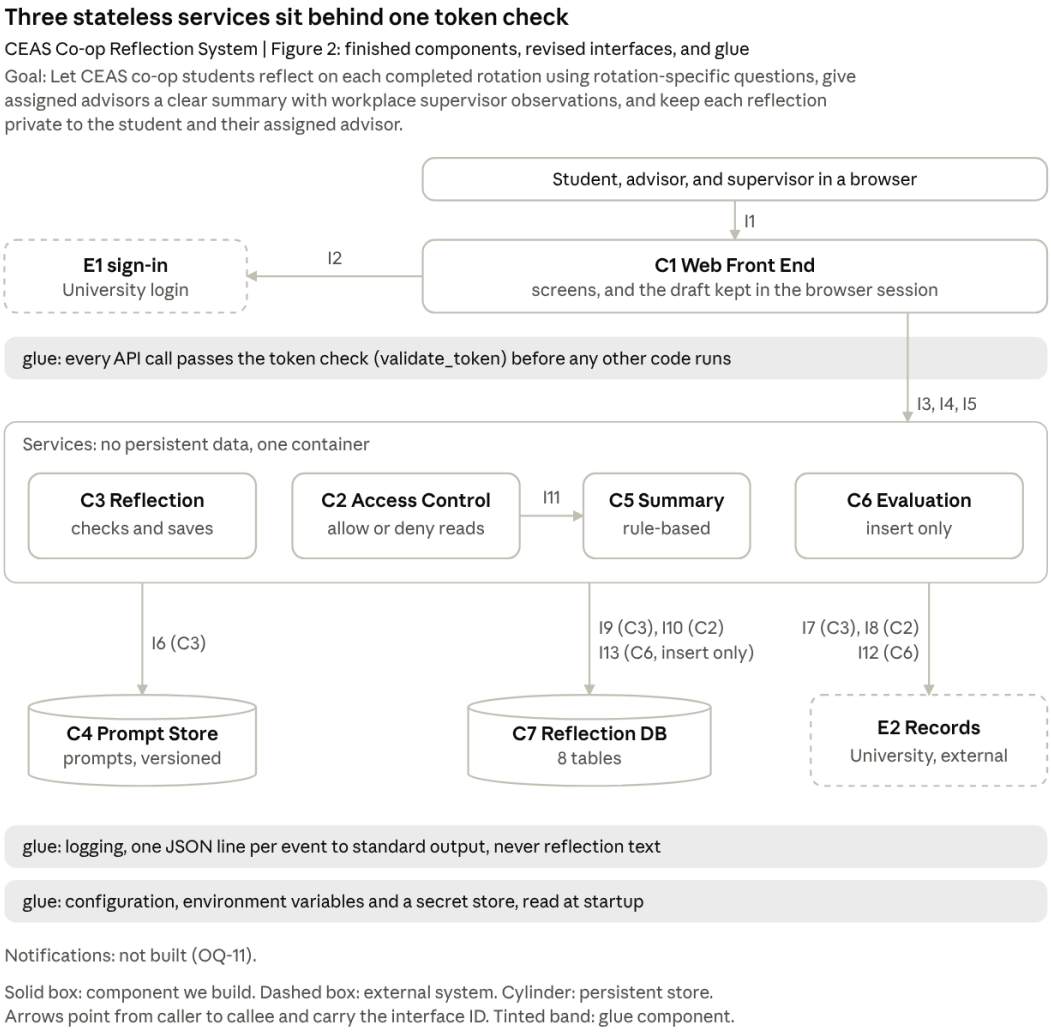


Figure 2 · block diagram, 7 components, 2 external systems, 3 glue bands

Figure 2 shows the finished system. C1 reaches the three services through the token check, the services reach C4, C7, and E2, and C5 is called only by C2. Arrows are bundled by caller, so the owner of each interface is written beside it.

Revised D0 interfaces

Interface	D0 version	Problem	D2 version
I3 (C1 to C3)	One payload with an "action" flag, a rotation_number, and the responses	One shape does three jobs (look up, check, submit), and the browser decides the rotation number	Four endpoints: form, acknowledge notice, check, submit. C3 picks the rotation number and its source from E2. The

Interface	D0 version	Problem	D2 version
			browser never sends rotation_source
I4 (C1 to C2)	"User identity and requested reflection"	Identity in the request can be forged, and C1 would need a second call to C5 for the summary	Identity comes only from the sign-in token (as in D1). GET reflection takes ?include=summary, so C1 never calls C5 and I11 stays inside C2
I5 (C1 to C6)	"Student/placement information and supervisor evaluation"	The browser names the student, and the parameter list is open ended	One endpoint with placement_id in the path and three body fields. C6 learns the student and rotation facts from E2

The other D0 interfaces keep their meaning. The I8 and I9 labels on the D0 diagram are swapped (see the corrections in the first section).

C4 Prompt Configuration Store

Data model. C4 has its own schema in the PostgreSQL server. C3 reads it with a read-only database role, and only the admin load script writes to it.

Entity	Key attributes	Relationship
PROMPT_SET	rotation_number (PK, integer), current_version (integer)	One row per supported rotation number. Each version of a set has 1..* PROMPT rows
PROMPT	prompt_id (PK, string, 1 to 32 characters), rotation_number (FK), set_version (integer), category (accomplishment, challenge, or goal), text (string), display_order (integer), required (boolean)	Each belongs to exactly one set. UQ (rotation_number, set_version, display_order)

Prompt rows are never edited. A wording change is a new set_version with new prompt_ids, so an old reflection still points at the prompts the student saw (R5). A valid version has at least one prompt in each of the three categories.

Behavior.

1. C3 loads all prompt sets into memory at startup and refreshes them every 10 minutes (setting PROMPT_CACHE_MINUTES).
2. If the rotation number has no PROMPT_SET row, return unsupported_rotation.
3. Pick the requested version, or current_version if none was asked for. If that version does not exist, return version_not_found.

4. Return the prompts in display_order. If the version lacks a category or any text is empty, return config_incomplete instead, because D0 says an incomplete prompt set is never shown.

The admin load script checks a whole new version first. It then adds the rows and moves current_version in one transaction, so a bad file changes nothing.

Interface (I6, in-process calls from C3).

Call	Inputs	Output
get_prompt_set	rotation_number (integer, 1 or more, required), version (integer, 1 or more, optional)	rotation_number, version, and prompts: an array of {prompt_id: string, category: string, text: string, required: boolean, display_order: integer}
list_supported_rotations	none	array of integers

Error code	When it happens
unsupported_rotation	No set exists for that rotation number
version_not_found	The requested version does not exist
config_incomplete	The version misses a category or has empty text
store_unavailable	The database cannot be reached and the cache is empty

C5 Summary Generator

Data model. No persistent data. A summary is built when it is asked for and never stored, because a stored copy would be a second copy of private text.

Behavior. C5 is rule-based and uses no AI model (R3). Its input is the reflection's answers in display order, plus the supervisor evaluation if there is one.

1. For each category in the order accomplishment, challenge, goal, collect the answers in that category and join them with a space.
2. If a category has no answer, the field is "No response recorded."
3. If a field is longer than 300 characters, cut it at the last sentence end (a period, question mark, or exclamation mark) at or before character 300. If there is none, cut at the last space before character 300 and add "...". Mark that field as truncated.
4. Never reword, paraphrase, or add words. The summary may only contain text the student wrote (US-02 INVEST check).
5. If an evaluation is given, copy its two fields with the same cut rule. Otherwise supervisor_observations is null.

The work is one pass over the text, so the cost is proportional to the text length and takes microseconds. Two requests for the same reflection give the same answer, so concurrent calls cannot disagree.

Interface (I11, called only by C2).

Call	Inputs	Output
summarize	responses: array of {prompt_id: string, category: string, response_text: string}, required, 3 to 20 items. evaluation: object with accomplishments and improvement_areas (strings), or null	accomplishment, challenge, goal (strings, up to 303 characters each), truncated (an object with three booleans), supervisor_observations (object with two strings, or null)

Error code	When it happens
invalid_input	An answer has no category or its text is not a string
summary_failed	Any other failure in C5. C2 then returns the original reflection with summary_status set to unavailable (D0 I11)

C1 receives the result through C2: GET /v1/reflections/{reflection_id}?include=summary adds a summary object and a summary_status field ("ok" or "unavailable") to the D1 response. Adding optional fields is not a breaking change under the D1 versioning rule.

C3 Reflection Service

Data model. No persistent data. The student's draft stays in C1's session, so an incomplete reflection never reaches the database (the D0 rule on the C3 to C7 link). C3 reads C4 and E2 and saves through C7's save_reflection (D1). On a student's first start, C3 also adds the STUDENT and PLACEMENT rows from the E2 record if they are missing, and it writes the notice acknowledgment.

Behavior. Figure 1 shows the student's whole flow, including the error paths. These are the steps C3 runs.

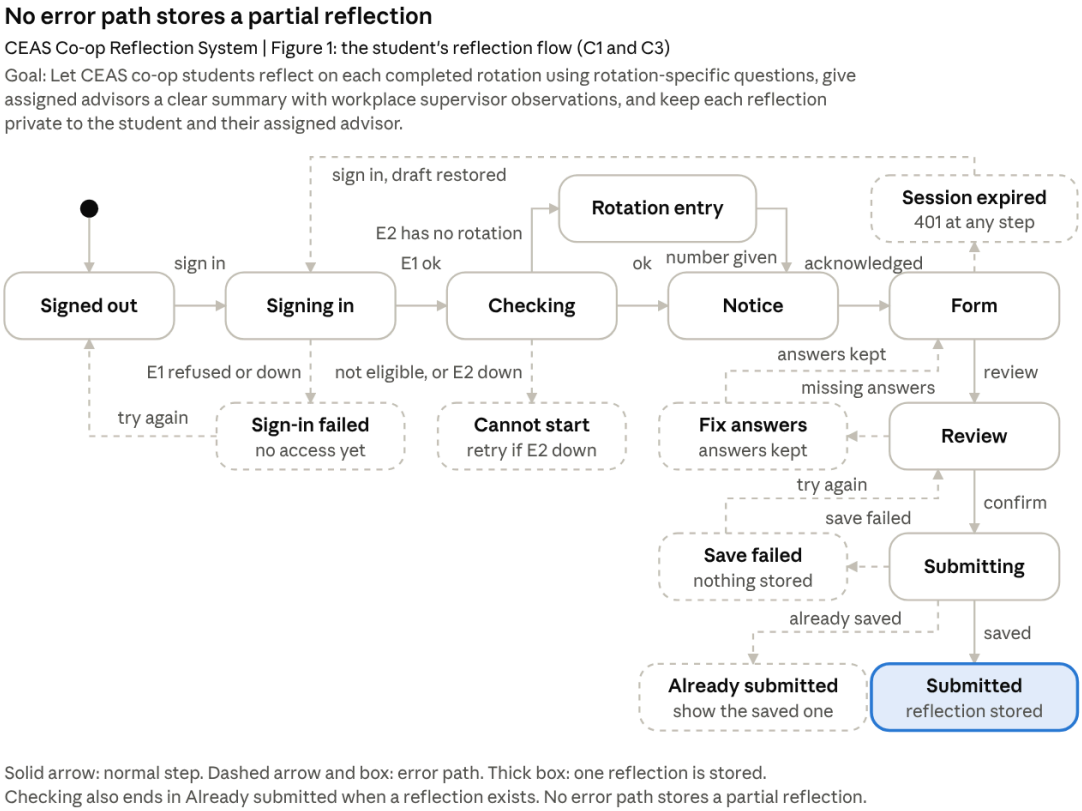


Figure 1 · student reflection flow, 8 states on the main path, 7 error or alternate states

Read Figure 1 along the top row, then down the right edge to Submitted. Every dashed box is a failure that either keeps the student's answers or stores nothing, so a partial reflection never exists.

1. **Start.** Check the sign-in token and that the caller is a student. Ask E2 (I7) for eligibility, the placement, and the rotation number. If the student is not eligible or the placement is not theirs, stop with not_eligible. If E2 cannot confirm, stop with records_unavailable, because D0 says the reflection cannot be started.
2. **Notice.** If the student has not acknowledged the current notice version, the response says so. C1 shows the notice first, and submit refuses until it is acknowledged (UC-01 precondition).
3. **Duplicate check.** If a reflection already exists for this placement and rotation, return reflection_already_submitted with its ID.
4. **Rotation.** Use E2's rotation number. If E2 has none, return rotation_number_needed. When the student supplies a supported number, use it and mark the source student_supplied (UC-01 A1).
5. **Prompts.** Get the prompt set from C4 (I6) and return the form.
6. **Check.** Compare the answers with the prompt set version the form carried, and list every problem at once (AC-01.4).
7. **Submit.** Run every check again, add each answer's category from the prompt set, call save_reflection (I9), and map its error codes to the errors below.

Interface (revised I3). Four endpoints. All need the sign-in token. Errors use the D1 body shape, with an optional details object: {"error": {"code": string, "message": string, "details": object}}.

GET /v1/placements/{placement_id} /reflection-form	Specification
Inputs	placement_id (path, string, 1 to 32 characters, required). rotation_number (query, integer, 1 or more, optional, used only when E2 has none)
200 output	placement_id (string), rotation_number (integer), rotation_source ("university" or "student_supplied"), prompt_set_version (integer), prompts (array of {prompt_id: string, category: string, text: string, required: boolean}), notice ({version: integer, acknowledged: boolean})
Errors	401 not_signed_in. 403 not_eligible (not a student, not eligible, or not this student's placement, same body for all). 409 reflection_already_submitted (details.reflection_id). 422 rotation_number_needed. 422 unsupported_rotation. 503 records_unavailable (E2 cannot confirm). 503 prompts_unavailable (C4 unreadable and cache empty). 500 internal_error

POST /v1/notice-acknowledgments	Specification
Inputs	notice_version (body, integer, 1 or more, required)
Output	201 with acknowledged_at (ISO 8601 UTC). A repeat call returns 200 with the first time
Errors	400 invalid_request. 401 not_signed_in. 422 unknown_notice_version (not the current version). 503 save_failed

POST /v1/placements/{placement_id}/reflection-checks	Specification
Inputs	placement_id (path, as above). Body: rotation_number (integer, 1 or more, required), prompt_set_version (integer, 1 or more, required), responses (array of 1 to 20 items, required, each {prompt_id: string 1 to 32 characters, text: string})
200 output	valid (boolean, always true). Nothing is stored

POST /v1/placements/{placement_id}/reflection-checks	Specification
Errors	401 not_signed_in. 403 not_eligible. 422 invalid_response (details: missing_prompt_ids, unknown_prompt_ids, too_long_prompt_ids, each an array that may be empty). 422 unsupported_rotation. 422 prompt_set_not_found. 503 records_unavailable

POST /v1/placements/{placement_id}/reflections	Specification
Inputs	Same as the check endpoint. If E2 has a rotation number, C3 uses it and ignores the one in the body
201 output	reflection_id (string, UUID), submitted_at (string, ISO 8601 UTC)
Errors	400 invalid_request. 401 not_signed_in. 403 not_eligible. 409 reflection_already_submitted (details.reflection_id). 422 invalid_response. 422 unsupported_rotation. 422 notice_required. 503 records_unavailable. 503 save_failed (database failure, nothing stored). 500 internal_error

C1 Web Front End

Data model. No persistent server data. The browser keeps two things: the draft answers in sessionStorage (one entry per placement, cleared on submit or sign-out), and the sign-in token in a secure, HTTP-only cookie that page scripts cannot read.

Behavior. Figure 1 is the student's flow. The other two roles follow short step lists.

- **Advisor:** After sign-in, C1 calls GET /v1/me, then GET /v1/advisors/me/students for the roster, then GET /v1/students/{student_id}/reflections for one student's list, then GET /v1/reflections/{reflection_id}?include=summary to show a reflection.
- **Supervisor:** The supervisor arrives at /evaluate/{placement_id} from CEAS (OQ-1), signs in, fills in the two fields, and submits to C6. The typed text stays on the page until it is saved.

D1 left out one thing the advisor flow needs, a way to find a student. D2 adds it to C2:

GET /v1/advisors/me/students (C2, called by C1)	Specification
Inputs	limit (query, integer, 1 to 50, default 20, optional). offset (query, integer, 0 or more, default 0, optional). Authorization header (required)
200 output	total (integer), roster_stale (boolean), checked_at (string, ISO 8601 UTC, when the list was last confirmed with E2), students (array of {student_id: UUID, display_name: string, reflection_count: integer}, sorted by display_name)
Behavior	Lists students with a current ADVISOR_ASSIGNMENT row for the caller. If the newest checked_at is older than 15 minutes, C2 refreshes from E2. If E2 cannot be reached, it returns the local rows with roster_stale true. The list holds names only, and opening a reflection still runs the full check in D1 Algorithm 1
Errors	400 invalid_request. 401 not_signed_in. 403 not_an_advisor (no ADVISOR row for the caller). 503 backend_unavailable

Interface. C1 is the caller, so its interface is what each screen calls and what it shows when the call fails.

Screen	Calls	When the call fails, C1 shows
Sign-in	GET /auth/login	"University sign-in is not available. Try again in a few minutes." No access until sign-in succeeds (D0 I2)
Notice	POST /v1/notice-acknowledgments	"We could not save your acknowledgement. Try again." The form stays locked
Reflection form	reflection-form	not_eligible: "You cannot reflect on this placement." records_unavailable: "We cannot confirm your enrollment right now. Try again later," with a retry button. rotation_number_needed: asks for a supported rotation number. reflection_already_submitted: opens that reflection
Review	reflection-checks	invalid_response: marks every unanswered prompt and returns to the form with the answers kept
Submit	reflections	save_failed or 503: "Your reflection was not saved. Try again," answers kept. 409: an "already submitted" page. 401: sign in again, then restore the draft

Screen	Calls	When the call fails, C1 shows
Advisor roster, list, and view	the three advisor GETs	403: "You do not have access to this reflection," with no content. 503: try again. summary_status unavailable: shows the reflection with a note that the summary is missing
Supervisor form	evaluations	invalid_evaluation: marks the fields. not_assigned_supervisor: "We could not match you to this placement." records_unavailable: retry with the text kept. duplicate_evaluation: "An evaluation was already submitted"

C6 Supervisor Evaluation Service

Data model. No persistent data of its own. C6 writes SUPERVISOR_EVALUATION (D1) through a database role that can insert and nothing else, so a faulty C6 cannot read any reflection or evaluation.

Behavior.

1. Check the sign-in token. The supervisor's identity becomes supervisor_ref (OQ-1).
2. Check the body: accomplishments and improvement_areas must each be 1 to 5,000 characters with at least one non-whitespace character.
3. Ask E2 (I12) whether this supervisor is assigned to this placement and rotation. If not, return not_assigned_supervisor. If E2 cannot confirm, return records_unavailable, because D0 says the evaluation is then not accepted.
4. If the placement has not started, return placement_not_started (OQ-10).
5. Insert the evaluation. If the unique pair (placement_id, rotation_number) is already taken, return duplicate_evaluation. Any other database failure returns save_failed and stores nothing.
6. Return the new evaluation_id.

Interface (revised I5). One endpoint. There is no read endpoint, so a supervisor cannot see an evaluation after sending it (R8).

POST /v1/placements/{placement_id}/evaluations	Specification
Inputs	placement_id (path, string, 1 to 32 characters, required). Body: rotation_number (integer, 1 or more, required), accomplishments (string, 1 to 5,000 characters, required), improvement_areas (string, 1 to 5,000 characters, required)
201 output	evaluation_id (string, UUID), submitted_at (string, ISO 8601 UTC)
Errors	400 invalid_request. 401 not_signed_in. 403 not_assigned_supervisor. 409 duplicate_evaluation. 422 invalid_evaluation (details.fields lists the bad field names). 422

POST /v1/placements/{placement_id}/evaluations	Specification
	placement_not_started. 503 records_unavailable. 503 save_failed. 500 internal_error

Glue components

Four pieces hold the system together: sign-in and the token check, logging, configuration, and notifications.

Sign-in and token check (E1)

Data model. No persistent data. The token is stateless. STUDENT and ADVISOR rows in C7 are created at a person's first sign-in from E2 data (step 5 below).

Behavior.

1. GET /auth/login sends the browser to E1 using the OpenID Connect authorization code flow, with a random state value.
2. GET /auth/callback checks the state value, exchanges the code with E1, and checks the ID token's signature, issuer, audience, and expiry. It allows 60 seconds of clock skew.
3. It sets the token cookie and sends the browser back to the page it came from.
4. Every API call passes through validate_token before any other code runs.
5. GET /v1/me finds the person's roles in the STUDENT and ADVISOR tables. If neither row exists, it asks E2 once and creates the matching row. If E2 knows nobody by that ID, the roles list is empty and that person can read nothing (US-04).

Call	Inputs	Output	Errors
GET /auth/login	return_to (query, string, a path on our site, optional)	302 redirect to E1	400 invalid_request (return_to points off our site)
GET /auth/callback	code and state (query, strings, required)	302 redirect, token cookie set	400 invalid_state. 403 sign_in_denied (E1 refused). 502 sign_in_unavailable (E1 unreachable)
POST /auth/logout	none	204, cookie cleared	none
validate_token (in-process)	the token cookie	university_id (string, 1 to 64 characters), display_name (string)	invalid_token (missing, altered, or wrong audience) and expired_token. Both

Call	Inputs	Output	Errors
			reach the caller as 401 not_signed_in
GET /v1/me	the token cookie	roles (array of "student" and "advisor", may be empty), student_id (UUID, only for students), advisor_id (UUID, only for advisors), display_name (string)	401 not_signed_in. 503 records_unavailable (first-time user and E2 unreachable)

Logging

Data model. No persistent data in the database. One JSON line per event goes to standard output, and the hosting platform keeps the lines.

Behavior. log_event writes one line and accepts only these fields: request_id, actor_university_id, reflection_id, student_id, placement_id, decision, reason, status, duration_ms, error_code. Reflection text, evaluation text, tokens, and cookies are never logged. The events are access_decision, reflection_saved, evaluation_saved, e2_call_failed, prompt_cache_refresh_failed, token_rejected, config_error, and unhandled_error. An identical error that repeats is written at most once every 10 seconds with a count.

Call	Inputs	Output	Errors
log_event	level ("info", "warn", or "error"), component (C1 to C7, "auth", or "config"), event (one of the names above), fields (an object limited to the allowed names, string values cut at 200 characters)	none	None reach the caller. If the write fails, the line goes to standard error and the request carries on

Configuration

Data model. No persistent data. Settings come from environment variables, and secrets come from the platform's secret store. Prompt wording is data in C4, not a setting.

Behavior. At startup the service loads every setting and checks it. A missing or out-of-range value stops the start with a message that names the variable. Nothing reloads while the service runs, so a change needs a restart.

Setting	Type and range	Default
E2_BASE_URL	HTTPS URL	none, required

Setting	Type and range	Default
E2_TIMEOUT_SECONDS	integer, 1 to 5	2
ASSIGNMENT_FRESHNESS_MINUTES	integer, 1 to 60	15
PROMPT_CACHE_MINUTES	integer, 1 to 60	10
MAX_RESPONSE_CHARS	integer, 100 to 20,000	5,000
SUMMARY_FIELD_CHARS	integer, 100 to 1,000	300
NOTICE_VERSION	integer, 1 or more	none, required
TOKEN_SKEW_SECONDS	integer, 0 to 120	60
DATABASE_URL, E1_CLIENT_ID, E1_CLIENT_SECRET	secret strings	none, required, from the secret store

The only call is load_settings at startup. Its errors are config_missing and config_out_of_range, and either one stops the start.

Notifications

Not built. No requirement asks for email or messages, D0 has no notification component, and an email would copy private text outside the system. If CEAS asks for them (OQ-11), a later component can read the reflection_saved and evaluation_saved log events, so C3 and C6 do not change. There is no data model, behavior, or interface to define.

Edge cases and error handling

Each of the seven components and three glue pieces has one row in each of the four categories: Boundary, Timing, Bad input, and External. That makes 40 rows (EC-01 to EC-40). Every decision starts with its kind (Retry, Degrade, Fall back, or Surface error) and names a handler. Figure 1 shows the error paths of the student flow. Rows for C2 and C7 repeat the key cases from D1 so the table is complete on its own.

ID	Component	Category	Edge case	Design decision	Handler and recoverable
EC-01	C1	Boundary	An answer is exactly 5,000 characters, 5,001 characters, or one non-space character.	Surface error at 5,001: a live character count blocks the field. 1 to 5,000 characters are accepted.	C1 form check, then C3 check. Recoverable: yes

ID	Component	Category	Edge case	Design decision	Handler and recoverable
EC-02	C1	Timing	The token expires while a student is typing a long answer.	Fall back: C1 saves the draft in sessionStorage, sends the student to sign in, and restores the draft.	C1 session guard. Recoverable: yes
EC-03	C1	Bad input	An answer is only spaces or line breaks, or one very long unbroken string.	Surface error: whitespace-only counts as empty (AC-01.4). A long string is accepted up to 5,000 characters.	C1 trim check, then C3. Recoverable: yes
EC-04	C1	External	C2, C3, or C6 is unreachable or returns 503.	Retry (manual): show "try again" and keep the draft. C1 never resends a submit by itself.	C1 error banner. Recoverable: yes
EC-05	C2	Boundary	An assignment's end_date is today, or its start_date is tomorrow.	Surface error for a future start (403 access_denied). An end_date of today still allows access (D1 Algorithm 1).	C2 access check. Recoverable: not applicable, it is the rule
EC-06	C2	Timing	An advisor is unassigned in E2 while the local row is under 15 minutes old.	Degrade: access continues until the row is 15 minutes old. The next read then asks E2 (R2).	C2 freshness check. Recoverable: yes, it corrects itself
EC-07	C2	Bad input	A malformed UUID, a missing token, or a signed-in user with no role.	Surface error: 400 invalid_request, 401 not_signed_in, or 403 access_denied, with no content.	C2 request check. Recoverable: yes

ID	Component	Category	Edge case	Design decision	Handler and recoverable
EC-08	C2	External	E2 is slow or down during an advisor check.	Fall back: use a fresh local row. Otherwise deny with 403 after a 2-second timeout. There is no automatic retry on a read.	C2 E2 client. Recoverable: yes, the user retries later
EC-09	C3	Boundary	rotation_number is 0, negative, or above the supported range.	Surface error: 422 unsupported_rotation.	C3 rotation check against C4. Recoverable: yes
EC-10	C3	Timing	Two submits at the same moment (a double click or two tabs).	Surface error: the database accepts one and the other gets 409 reflection_already_submitted with the saved ID (D1 Algorithm 2). C1 also disables the button while sending.	C7 unique pair, mapped by C3. Recoverable: yes, nothing is lost
EC-11	C3	Bad input	A missing, unknown, or repeated prompt_id, or a rotation_source field in the body.	Surface error: 422 invalid_response lists every problem prompt. An unknown body field gets 400 invalid_request.	C3 request validator. Recoverable: yes
EC-12	C3	External	E2 cannot confirm eligibility, or has no rotation number.	Surface error: 503 records_unavailable, and the reflection cannot start (D0). With no rotation number, 422 rotation_number_	C3 start handler. Recoverable: yes

ID	Component	Category	Edge case	Design decision	Handler and recoverable
				needed lets the student supply one (UC-01 A1).	
EC-13	C4	Boundary	A rotation has zero prompts, or a set misses one of the three categories.	Surface error: config_incomplete. C3 treats the rotation as unsupported and never shows a partial prompt set.	C4 set check. Recoverable: yes, after the data is fixed
EC-14	C4	Timing	An admin loads a new prompt version while students are mid-reflection.	Degrade: old versions stay. A form keeps the version it loaded, and the check uses that version (R5).	C4 versioning, C3 check. Recoverable: yes, no impact
EC-15	C4	Bad input	A load file has a duplicate prompt_id, an unknown category, or empty text.	Surface error: the script rejects the whole file and the live version stays.	C4 admin load script. Recoverable: yes, fix and reload
EC-16	C4	External	The database that holds C4 is unreachable.	Fall back: C3 serves its in-memory copy of the last good prompt sets. With an empty cache, 503 prompts_unavailable.	C3 prompt client. Recoverable: yes
EC-17	C5	Boundary	An answer is over 300 characters, or is one long sentence with no end mark.	Degrade: cut at the last sentence end within 300 characters, else at the last space plus "...", and set the truncated flag.	C5 cut rule. Recoverable: not applicable, it is the rule
EC-18	C5	Timing	Two summaries of the same reflection	Retry is safe: C5 keeps no state and	C5, none needed.

ID	Component	Category	Edge case	Design decision	Handler and recoverable
			are requested at once.	gives the same answer every time, so no coordination is needed.	Recoverable: not applicable
EC-19	C5	Bad input	A category has no answer, or an answer has no category.	Surface error: no answer gives "No response recorded." An answer with no category gives invalid_input. Nothing is invented.	C5 builder. Recoverable: yes for the first, a caller bug for the second
EC-20	C5	External	C5 fails or takes longer than 1 second.	Degrade: C2 returns the original reflection with summary_status "unavailable" (D0 I11) and logs the failure.	C2 I11 client. Recoverable: yes
EC-21	C6	Boundary	An evaluation field has 1 character or 5,001, or the placement has not started.	Surface error: 1 to 5,000 are accepted. Others get 422 invalid_evaluation. A placement that has not started gets 422 placement_not_started.	C6 validator. Recoverable: yes
EC-22	C6	Timing	The same supervisor submits twice (a double click or two tabs).	Surface error: 409 duplicate_evaluation. The first one stands, and there is no edit path (OQ-3).	C7 unique pair, mapped by C6. Recoverable: yes, corrections go through an admin

ID	Component	Category	Edge case	Design decision	Handler and recoverable
EC-23	C6	Bad input	A field is missing or whitespace only, or the placement_id is unknown or not theirs.	Surface error: 422 invalid_evaluation names the fields. An unknown or unassigned placement gets the same 403 not_assigned_supervisor, so nothing leaks.	C6 validator. Recoverable: yes
EC-24	C6	External	E2 cannot confirm the supervisor's assignment, or the insert fails.	Surface error: 503 records_unavailable and nothing is stored (D0). A failed insert gives 503 save_failed. C1 keeps the typed text.	C6 E2 client and insert handler. Recoverable: yes
EC-25	C7	Boundary	A save with 2 or 21 answers, or an answer over 5,000 characters.	Surface error: invalid_response. A CHECK constraint on length and non-whitespace backs this up in the database.	save_reflection validator and C7 CHECK. Recoverable: yes
EC-26	C7	Timing	Two saves race for one placement and rotation, or a save runs longer than 5 seconds.	Surface error: the unique pair rolls back the loser (duplicate_reflection). A 5-second statement timeout rolls back with save_failed.	C7 constraint and timeout. Recoverable: yes
EC-27	C7	Bad input	A placement that is not the student's, an unsupported rotation, or null fields.	Surface error: not_authorized, invalid_rotation, or invalid_response. Nothing is stored.	save_reflection validator. Recoverable: yes

ID	Component	Category	Edge case	Design decision	Handler and recoverable
EC-28	C7	External	The database is down, the disk is full, or connections run out.	Surface error: reads return 503 backend_unavailable and saves roll back as save_failed. The platform health check restarts the container, and recovery uses the daily backup (assumption: up to 24 hours of loss).	C7 connection layer and platform. Recoverable: yes once the database is back
EC-29	Sign-in	Boundary	A token is used in the exact second it expires, or the clocks differ.	Fall back: allow 60 seconds of skew. After that, surface 401 not_signed_in.	validate_token. Recoverable: yes, sign in again
EC-30	Sign-in	Timing	One person signs in on two devices, or the token expires during a submit.	Surface error: each device has its own token. An expiry during a submit gives 401 and no partial save, because the save is one transaction. C1 restores the draft (EC-02).	validate_token and C1. Recoverable: yes
EC-31	Sign-in	Bad input	A forged, altered, or wrong-audience token, or a bad state value on the callback.	Surface error: 401 not_signed_in, or 400 invalid_state. The event is logged without the token.	validate_token and /auth/callback. Recoverable: yes
EC-32	Sign-in	External	E1 is down or refuses the sign-in.	Surface error: 502 sign_in_unavailable or 403 sign_in_denied. People already signed in keep working until their	/auth/callback and the C1 sign-in screen. Recoverable: yes when E1 is back

ID	Component	Category	Edge case	Design decision	Handler and recoverable
				token expires. There is no password fallback (D0 decision log).	
EC-33	Logging	Boundary	A field value is very long, or reflection text is passed in by mistake.	Degrade: only allowed field names are written, values are cut at 200 characters, and no text field is an allowed name.	log_event allowlist. Recoverable: not applicable
EC-34	Logging	Timing	An E2 outage produces thousands of identical errors.	Degrade: an identical error is written once every 10 seconds, with a count.	log_event rate limit. Recoverable: yes
EC-35	Logging	Bad input	A caller uses an unknown event or field name.	Degrade: the unknown field is dropped and a log_field_rejected line is written.	log_event allowlist. Recoverable: yes
EC-36	Logging	External	The log destination is unavailable.	Fall back: write to standard error and keep serving. A logging failure never fails a request.	log_event. Recoverable: yes
EC-37	Configuration	Boundary	ASSIGNMENT_FRESHNESS_MINUTES is set to 0 or 100,000.	Surface error: the service refuses to start (config_out_of_range). The allowed range is 1 to 60.	load_settings. Recoverable: yes, fix the value
EC-38	Configuration	Timing	A setting is changed while the service runs.	Fall back: keep the old values until the next restart. Requests in	load_settings and the platform restart.

ID	Component	Category	Edge case	Design decision	Handler and recoverable
				progress finish normally.	Recoverable: yes
EC-39	Configuration	Bad input	A required setting is missing or malformed, for example an empty E2_BASE_URL.	Surface error: the service stops at startup with a message that names the variable (config_missing).	load_settings. Recoverable: yes
EC-40	Configuration	External	The secret store is unavailable at startup.	Fall back: the new instance does not start and the platform retries, while any running instance keeps serving.	load_settings and the platform. Recoverable: yes

Requirements traceability table

Every requirement from Assignment 4 maps to named design elements below: 4 user stories, 2 use cases, 7 acceptance criteria, and the 6 draft rules, plus 5 non-functional requirements that D0 states or implies.

ID and requirement	Priority	Design elements	How it is satisfied	Edge cases and figures
US-01 Reflect on a completed rotation with rotation-specific questions	Must	PROMPT_SET and PROMPT (C4), the four C3 endpoints, REFLECTION and REFLECTION_RESPONSE (C7)	C4 holds the prompts per rotation, C3 checks that every category is answered, and C7 stores one reflection with its answers by category	EC-09 to EC-16, EC-25 to EC-27, Figure 1
US-02 Review reflection insights as a summary	Must	summarize (C5), GET /v1/reflections/{id}?include=summary, GET /v1/advisors/me/students (C2)	C5 copies each of the three fields (cut at 300 characters, no added words), and C2 returns the summary with the reflection	EC-17 to EC-20, Figure 2

ID and requirement	Priority	Design elements	How it is satisfied	Edge cases and figures
US-03 Submit workplace observations	Must	POST /v1/placements/{id}/evaluations (C6), SUPERVISOR_EVALUATION (C7), the E2 check (I12)	C6 verifies the assignment, inserts one evaluation, and C2 returns it to the assigned advisor in the evaluation field	EC-21 to EC-24
US-04 Restrict access to reflection content	Must	Algorithm 1 (C2), ADVISOR_ASSIGNMENT (C7), the token check, the identical 403 body, no key from SUPERVISOR_EVALUATION to REFLECTION, C6's insert-only role	Only the owner or the current advisor gets content, and content is never read before the decision	EC-05 to EC-08, EC-23, EC-30 to EC-32
UC-01 Complete a rotation-specific reflection (main flow, A1, E1)	Must	Figure 1 states, the four C3 endpoints, save_reflection	Each main step is a state in Figure 1. A1 is the Rotation entry state and E1 is the Fix answers state	EC-01 to EC-04, EC-09 to EC-12, Figure 1
UC-02 Read an authorized student reflection (main flow, A1, E1)	Must	GET /v1/advisors/me/students, GET /v1/students/{id}/reflections, GET /v1/reflections/{id} (C2), Algorithm 1	An advisor finds a student, then reads. A1 is the owner check and E1 is the denied response	EC-05 to EC-08
UC-01 and UC-02 preconditions (signed in, enrolled, notice acknowledged, assignments available)	Must	/auth endpoints and GET /v1/me, the E2 reads (I7, I8), ACKNOWLEDGMENT and POST /v1/notice-acknowledgments, ADVISOR_ASSIGNMENT	Sign-in sets identity, E2 confirms enrollment and assignments, and the notice is stored per version	EC-12, EC-29 to EC-32
AC-01.1 Rotation-specific prompts are presented	Must	get_prompt_set (C4), reflection-form 200 output (C3)	The form returns the prompt set for the recorded rotation, with	EC-09, EC-13, EC-14

ID and requirement	Priority	Design elements	How it is satisfied	Edge cases and figures
			a required flag on each prompt	
AC-01.2 Completed submission is stored once	Must	save_reflection (Algorithm 2), unique pair on REFLECTION, POST reflections 201	One transaction stores one reflection with its answers, student, placement, and rotation	EC-10, EC-26
AC-01.3 Student-supplied rotation is recorded as such	Must	rotation_number_needed (422), rotation_source on REFLECTION, Rotation entry state	C3 sets rotation_source to student_supplied on the server, so the browser cannot claim otherwise	EC-12, Figure 1
AC-01.4 Missing required responses block submission	Must	reflection-checks endpoint and its details lists, Fix answers state, draft in the session	Every unanswered prompt is listed, the other answers stay, and no reflection is created	EC-03, EC-11, EC-25, Figure 1
AC-02.1 Assigned advisor can read	Must	Algorithm 1 steps 4 and 5, ADVISOR_ASSIGNMENT	A current assignment row (or a fresh answer from E2) allows the read	EC-05, EC-06
AC-02.2 Student can read their own	Must	Algorithm 1 step 3, student_id on REFLECTION	One indexed comparison of the owner	EC-07
AC-02.3 Unauthorized access, including a direct ID, returns nothing	Must	Algorithm 1 steps 1 and 2, the identical 403 access_denied body	No content and the same response for an unknown ID	EC-07, EC-08
DR-1 Each supported rotation has a configured prompt set	Must	PROMPT_SET and PROMPT (C4), the admin load script	The mapping is data in C4, and tests seed it with known mappings. Real wording waits on OQ-4	EC-13 to EC-15
DR-2 Each prompt set requires an accomplishment, a challenge, and a	Must	The category rule in C4, the C3 check, the CHECK constraint in C7	Three layers check it: C4 at load, C3 at the check, C7 at the save	EC-13, EC-25

ID and requirement	Priority	Design elements	How it is satisfied	Edge cases and figures
goal, each with a non-whitespace character				
DR-3 A missing rotation number can be supplied by the student and is recorded as student-supplied	Must	rotation_source, the C3 start steps	Same mechanism as AC-01.3	EC-12
DR-4 Only the owner and the assigned advisor can read. Other students, unassigned advisors, supervisors, and IT staff cannot	Must	Algorithm 1, roles taken only from the STUDENT and ADVISOR tables	No other role exists in the design, so no other person can be allowed	EC-07
DR-5 Supervisor evaluations are separate from reflections, and submitting one grants no reflection access	Must	SUPERVISOR_EVALUATION with no key to REFLECTION, C6's insert-only role, no read endpoint in C6	The separation is in the table design and the database role, not only in code	EC-22 to EC-24
DR-6 Enrollment, advisor assignments, the data-use notice, and extra access roles must be confirmed. Retention and deletion need separate requirements	Should	The E2 reads, ACKNOWLEDGMENT. No design element for retention or deletion	Confirmations are covered by the E2 reads and the stored notice version. Retention and deletion are a stated gap: the university has not given guidance, so the system never deletes data and no deletion feature exists (OQ-5)	OQ-5, OQ-8
NFR-1 Each page load or submit	Should	Indexed lookups (D1 Algorithm 1), the	Database work takes milliseconds, and the	EC-08, EC-16

ID and requirement	Priority	Design elements	How it is satisfied	Edge cases and figures
finishes in under 2 seconds under normal load (D0 working target)		prompt cache, the 2-second E2 timeout, stateless services	E2 call is capped at 2 seconds. In an E2 failure the target can be missed, which we accept (R2)	
NFR-2 Works in an ordinary browser on a laptop or phone, with no special hardware (D0)	Should	C1 as a static React site with small pages, Figure 2	There is no hardware or native app. The pages are forms and short text	EC-04
NFR-3 No incomplete record is ever stored (D0 error handling)	Must	One transaction in save_reflection, the draft kept in the session, Figure 1	A save is all or nothing, and drafts never reach the database	EC-10, EC-26, EC-28
NFR-4 Reflection text never appears outside the allowed readers, including in logs and errors	Must	The logging allowlist, error bodies without text, no content read before allow	Text fields are not allowed log fields, and error bodies carry codes only	EC-33, EC-35
NFR-5 The system recovers from a failure without losing saved reflections	Should	Daily backup, the platform health check and restart	Assumption: a daily backup and automatic restart. No requirement states a recovery time, so the 24-hour figure is ours	EC-28

Reverse check: design elements with no requirement behind them

Design element	Requirement behind it	Verdict
C4 Prompt Configuration Store and its versioning	US-01, DR-1. Versioning keeps old reflections matching the prompts the student saw (R5)	Keep. Versioning is a design addition
C5 Summary Generator	US-02	Keep
C6 Supervisor Evaluation Service	US-03, DR-5	Keep

Design element	Requirement behind it	Verdict
GET /v1/advisors/me/students	UC-02 step 1, which assumes an advisor can find a student	Keep. Found as a D1 gap while writing D2
GET /v1/me and the /auth endpoints	UC-01 and UC-02 preconditions, DR-4 (roles)	Keep
ACKNOWLEDGMENT table and notice endpoint	UC-01 precondition, DR-6	Keep
Configuration glue	NFR-1 (timeouts and freshness are settings)	Keep
Logging glue	No requirement asks for logs. It supports NFR-4 and lets IT check US-04 afterward	Keep, but it is the closest thing to scope creep. Worth considering: cut it to access_decision and error events only
Notifications	None	Not built (OQ-11)

Gaps in this table

- **DR-6, retention and deletion:** no design yet, because the draft rules say guidance must come from the university first (OQ-5).
- **NFR-1, E2 failure case:** the 2-second target can be missed when E2 is slow. This is accepted, because denying access is safer than skipping the check.
- **NFR-5, recovery:** the backup and restart plan is an assumption, because no requirement names a recovery time.
- **US-03, supervisor sign-in:** the design works for any verified supervisor_ref, but how supervisors sign in is open (OQ-1).

Design rationale

Nine decisions shaped D1 and D2. The D0 decision log (university sign-in, a separate prompt store, one access service, separate supervisor evaluations, student-entered rotation) is not repeated here. The last column is for the peer review: none of these has been reviewed yet, so each says so.

ID	Decision	Changed by peer review?
R1	We chose a relational database (PostgreSQL) over a document store because unique constraints and one-transaction saves enforce our duplicate and all-or-nothing rules (D1). We accepted the trade-off of fixed tables and migration work because schema changes are rare and Alembic tracks them.	Not yet reviewed

ID	Decision	Changed by peer review?
R2	We chose a local copy of advisor assignments, refreshed from E2 every 15 minutes, over asking E2 on every read because every read would otherwise wait on E2 and fail when E2 is down (D1). We accepted the trade-off of up to 15 minutes of access after an advisor is unassigned because the window is a setting that CEAS can lower to 1 minute.	Not yet reviewed
R3	We chose a rule-based summary (C5) over an AI-written summary because a rule can only copy text the student wrote (US-02), keeps reflection text inside our system, and costs nothing. We accepted the trade-off of less fluent summaries, which are cut and not rewritten, because the advisor can read the full reflection beside the summary.	Not yet reviewed
R4	We chose to keep the student's draft in the browser session over saving drafts on the server because D0 says no incomplete record is stored, and every extra stored copy of private text is one more thing to protect. We accepted the trade-off of a lost draft when browser storage is cleared or the student changes device because a student can retype an answer, while a leaked draft cannot be taken back.	Not yet reviewed
R5	We chose prompts that are never edited and are versioned over editing prompts in place because an old reflection must still point at the prompts the student actually saw. We accepted the trade-off of more rows, with retired prompts staying in C4, because the extra storage is tiny.	Not yet reviewed
R6	We chose revised interfaces I3, I4, and I5 (four C3 endpoints, identity only from the token, one C6 endpoint) over the D0 versions because the browser cannot be trusted to name the student, the rotation source, or the user. We accepted the trade-off of more endpoints because each does one job and has its own errors. The D0 text is unchanged.	Not yet reviewed
R7	We chose one identical denied response for an unknown ID and a forbidden ID over separate 404 and 403 answers because AC-02.3 says a direct request by ID returns nothing, and a different answer would tell outsiders which IDs exist. We accepted the trade-off of harder debugging for real users because the true reason is written in the log.	Not yet reviewed
R8	We chose write-only access for supervisors over letting them view or edit what they sent because DR-5 says submitting grants no reflection access, and a read path is a new way in. We	Not yet reviewed

ID	Decision	Changed by peer review?
	accepted the trade-off that a supervisor cannot fix a mistake themselves because a CEAS administrator can correct it (OQ-3).	
R9	We chose stateless services with a token checked on every request over a server-side session table because there is then no session store to secure or lose. We accepted the trade-off that a token cannot be cancelled before it expires because E1 sets the expiry, which the team must confirm is short.	Not yet reviewed

Peer design review notes

To fill in after the review (delete this line then): the team has not been reviewed yet, so the action items below are empty rows. Use one row for each piece of feedback from each reviewing team. "We should look at error handling" is not an action item. A row needs a specific action, an owner, a priority, and a due date.

Pre-review audit of D0 (walked box by box)

D0 component	Detailed in	D0 owner	Status before review
C1 Web Front End	D2	Maria Malik	Done: states, screens, errors
C2 Access Control Service	D1 (plus the advisor roster endpoint in D2)	Hinna Parwez	Done
C3 Reflection Service	D2	Hinna Parwez	Done: four endpoints
C4 Prompt Configuration Store	D2	Priyanshi Kulshrestha	Done: wording waits on OQ-4
C5 Summary Generator	D2	Mbathio Lo	Done: rule-based
C6 Supervisor Evaluation Service	D2	Aradana Nair	Done: supervisor sign-in waits on OQ-1
C7 Reflection Database	D1	Priyanshi Kulshrestha	Done
Glue (sign-in, logging, configuration, notifications)	D2	Hinna Parwez	Done

Peer review we gave to ZNERV

ID	Rubric criterion	Issue we found	Suggested action	Priority
ZN-01	Feasibility	The team has no Electrical Engineering member, yet stimulation circuit decisions are central.	Name the faculty or specialist who will review the circuit, and say when.	Must-fix
ZN-02	Risk	Testing on people is not addressed. Human-subject approval and any university or facility approval for the device are missing.	Say whether testing will be on the team, on volunteers, or only on a bench model, and what approval is needed for each.	Must-fix
ZN-03	Risk	The description says safety monitoring and fault protection take priority, but it doesn't say what the limits are (maximum current, voltage, duration, or what triggers a stop).	Add the specific safety limits and the fault conditions that end a session.	Must-fix
ZN-04	Clarity	The constraints are one dense paragraph covering economic, professional, ethical, and social constraints.	Split it into four labeled short lists so a reader can find each category.	Worth considering

Action items

ID	Rubric criterion	Issue and team that raised it	Action	Owner	Priority	Due date
PR-01	Clarity	OQ-3, R8, and EC-22 disagree on how corrections work (team ZNERV)	Pick one correction process and use it in all three places	Priyanshi Kulshrestha	Must-fix	Oct 14

ID	Rubric criterion	Issue and team that raised it	Action	Owner	Priority	Due date
PR-02	Clarity	Students can't read the supervisor evaluation, but OQ-6 says they can (team ZNERV)	Add a student read path, or change OQ-6 to say students don't see it	Maria Malik	Must-fix	Oct 14
PR-03	Feasibility	E1 may not accept supervisor logins (OQ-1) (team ZNERV)	Name the supervisor sign-in mechanism and add a fallback	Aradana Nair	Must-fix	Oct 14
PR-04	Completeness	Figures 1 and 2 are described but not drawn (team ZNERV)	Draw both figures, with the project title and goal at the top of each	Mbathio Lo	Must-fix	Oct 14
PR-05	Completeness	Glue heading lists notifications, which isn't built, and the config and log lists are missing C5's timeout and log_field_rejected (team ZNERV)	Remove notifications from the glue heading, add C5_TIMEOUT_SECONDS to config, and add log_field_rejected to the log events	Hinna Parwez	Worth Considering	Oct 14

Summary

- **Criterion with the most feedback:** to fill in after the review.
- **Top two issues we raised while reviewing other teams:** to fill in after the review.